

Foundations AI Assignment 2B

(Data Science: Data Analysis and Modelling Assignment)

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Summary of data:

In response to the Harvard declaration regarding data scientists being “the sexist job of the 21st century” the dataset “2023-S2-FOAI-assignment.csv” has been by the CEO of a global data analytics company “In the Know”. The data they have collected includes information about salaries, employment data, demographic data, and company information in the data science industry from 2020 to 2023. As part of this an area that was collected was “work_year” which regards the year the salary was paid, “experience_level” the experience level in the job during the year there information was recorded. The different job titles were recorded in the column “job_title”. “In the Know” also collected different data about salary including data column “salary” which is the total gross salary amount they got paid and what currency they got paid in and also value converted into USD. The employee's primary country of residence for that work year is indicated, along with the extent of remote work undertaken, categorised into three levels: 0% remote, 50% partial remote, and 80%+ fully remote. They also gathered include in the data set the “company_location” which is the country of the employer's main office or contracting branch and the “company_size” the average number of people that worked for the company during the year categorised as small (less than 50 employees), medium (50-250 employees), or large (more than 250 employees) with S,M,L corresponding. As such this dataset provides a comprehensive overview of the employment and work arrangements recorded from 2020 to 2023.

The dataset contains 1500 rows, with 11 columns making a total of 16500 values, excluding those that are missing. Within we have many different data points collected about different aspects. Some of which are nominal where things such as company size are put into different groups or classes which have no particular order, we have interval ratio with salary where we can perform computations with it. Of the data the categorical variables there are experience_level, employment_type, job_title, salary_currency, employee_residence, company_location and company_size.

Statement of investigation:

This investigation seeks to understand the relationship between different aspects surrounding the data science industry in order to help make a just system using predictions that can assign individuals suitable and accurate rewards reflective of their individual circumstances and positions, including skills, experience, and job roles.

Data cleaning

- Outliers

Outliers: salary, salary in usd

By visualising the data into box plots some outliers were uncovered in both “salary” and “salary_in_USD ” using the sage maker system to remove rows, by setting the standard deviation at 2 any outliers were removed for the column “salary_in_USD”, however for the column salary since the values within are of different currency we cannot remove any of the values as the conversion rate could make them far higher and lower making the outliers bias against countries outside the USD which is the prominent currency.

- Missing values

First step was to check for missing values, using Sage maker identifying the missing values was simple. It was found that columns “salary” and “salary_in_USD” were the only ones with missing values both missing 6 values. As shown below. The steps undertaken to resolve this were first to fix the missing values in the “salary_in_USD” column, checking the visualisation at the top it was observed that the data was relatively skewed thus it was chosen to fill the missing values with the median. It was not possible to have checked the median or mean for “salary” as they were in different currencies making some far larger and smaller, thus “salary_in_USD” was chosen. The median value calculated was \$130000 and this value replaced all of the missing values. Since all the missing values where in salary were in USD we can then replace the missing values in “salary” with the same median value.

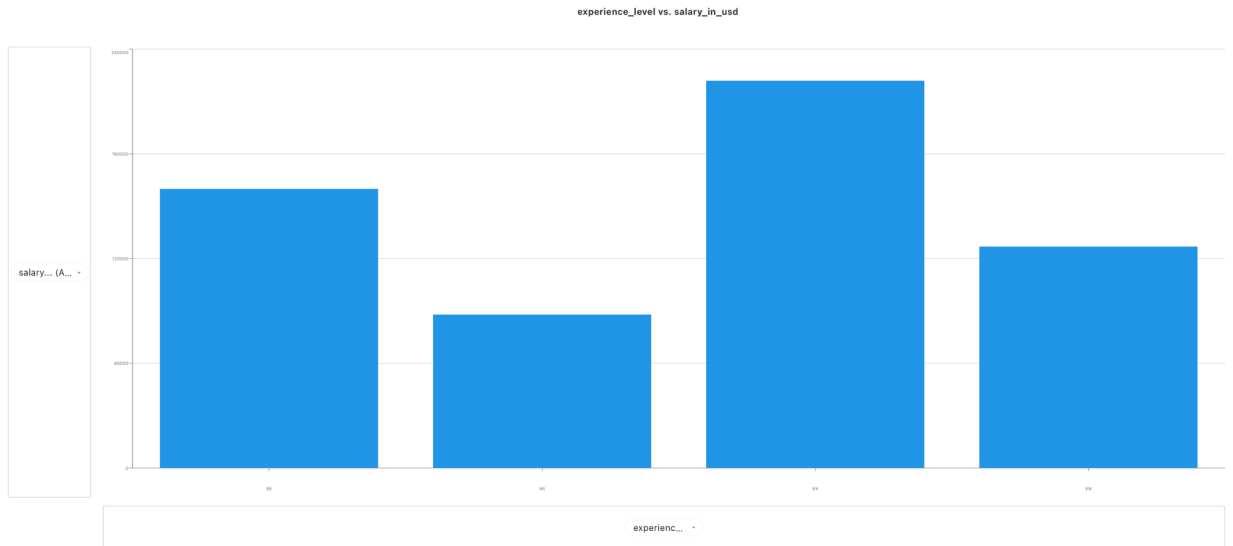
	Column name	Data type 	Feature type 	Missing 	Mismatched 	Unique 	Mode 
	salary	123 Numeric 	-	0.40% (6)	0.00% (0)	520	100,000
	salary_in_usd	123 Numeric 	-	0.40% (6)	0.00% (0)	656	100,000

Data exploration

- Charts

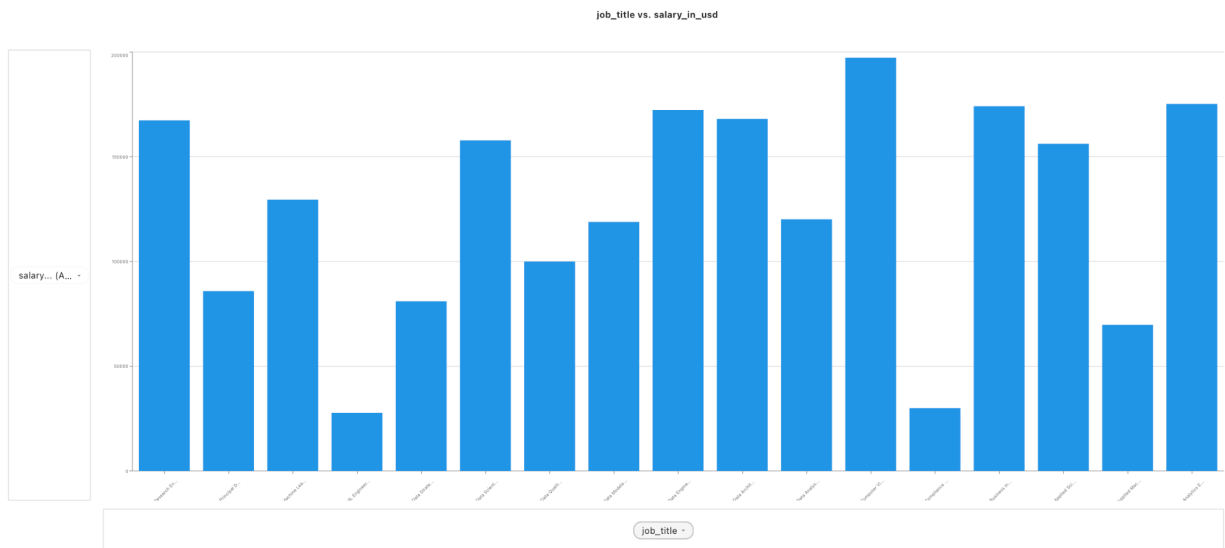
Bar Chart - Average Salary by Experience Level:

- This bar chart displays the average salary for different experience levels. The x-axis represents the various experience levels and the y-axis represents the average salary. Each bar corresponds to a different experience level, allowing viewers to compare how salaries vary across different stages of professional development. The graph below indicates that EX gets paid significantly more than the others and more than double MI which is lowest in terms of Average salary.



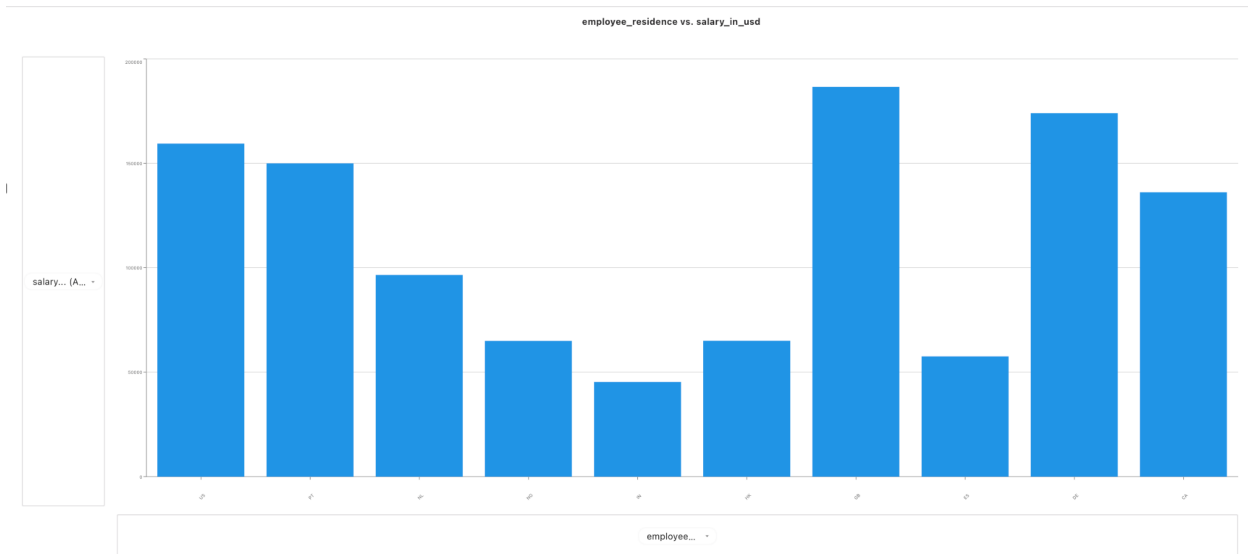
Bar Chart - Average Salary by Job Title:

- In this bar chart, the average salary for various job titles is depicted. Each job title corresponds to a bar on the x-axis, while the y-axis represents the average salary. By looking at the heights of the bars it can be seen, which job titles are getting paid the most. Some of the leading job titles by salary are Computer Vision Engineer, data engineer, business data analysts and analytics expert. On the lower side it is observed that compliance data analytic and ML engineers. As such the job titles and roles have an effect on salary.



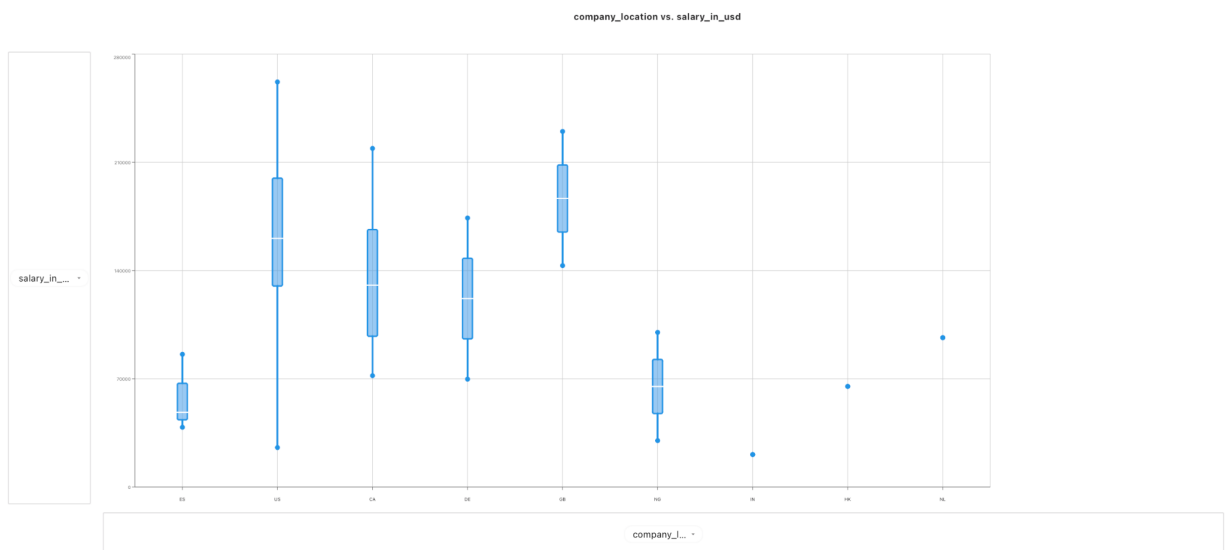
Bar Chart - Average Salary by Employee Residence:

- This bar chart showcases the average salary based on the primary country of residence of employees. Each respective country's name is plotted on the x-axis, while the average salary is represented on the y-axis. It can be seen that different employee residences have an impact on the salary as below there are different salary amounts, where some are far lower than others. Leading in salary is GB



Boxplot - Average Salary by Company Location:

- Below are box plots of average salary grouped by the company location, this is where the median, quartiles, and potential outliers within each are displayed. In the US it can be seen that there is a huge range and compared to the locations on the right where there are not many values recorded for are dots. The median values are generally close to each other, however the range has some great spread.

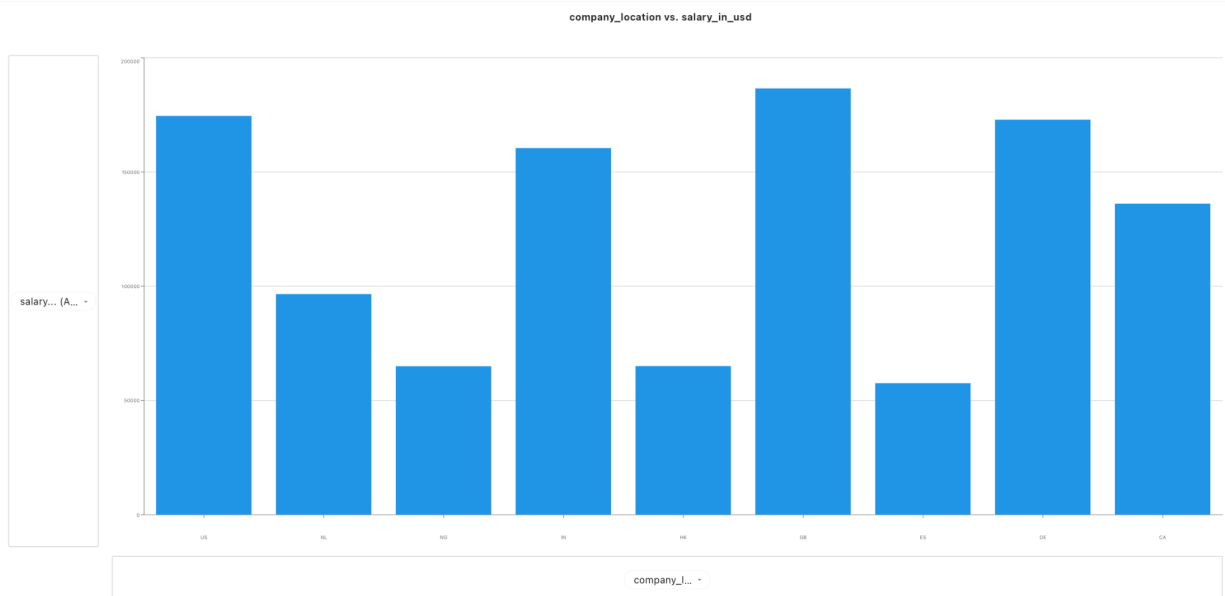


boxplot- Average Salary by Company Size:

- Similar to the previous boxplot, this visualisation compares the average salary, but this time based on the company's size (s, m, l). The x-axis represents company sizes, and the y-axis shows salary. The graph shows that medium sized companies have a higher pay on average going by the median and the higher inter quartile range. Small companies sit at the bottom with a very predictable and low salary, whilst large companies sit in between.



- The connection between the geographical location of a company and the average salary earned by its employees across various countries is shown below, using the bar chart we can see the average salary for each country separated in their own individual bins. Leading in salary is the company location of “GB” which is located in the United Kingdom, this is followed by DE or Delaware and “US” or the United States. It can be seen that depending on the company location individuals have a difference in pay, where depending on where one worked they could receive more than double elsewhere. If the salary average were on similar levels it could be assumed that company location has no impact on pay however we can reject that statement as there is a notable difference in average salary.

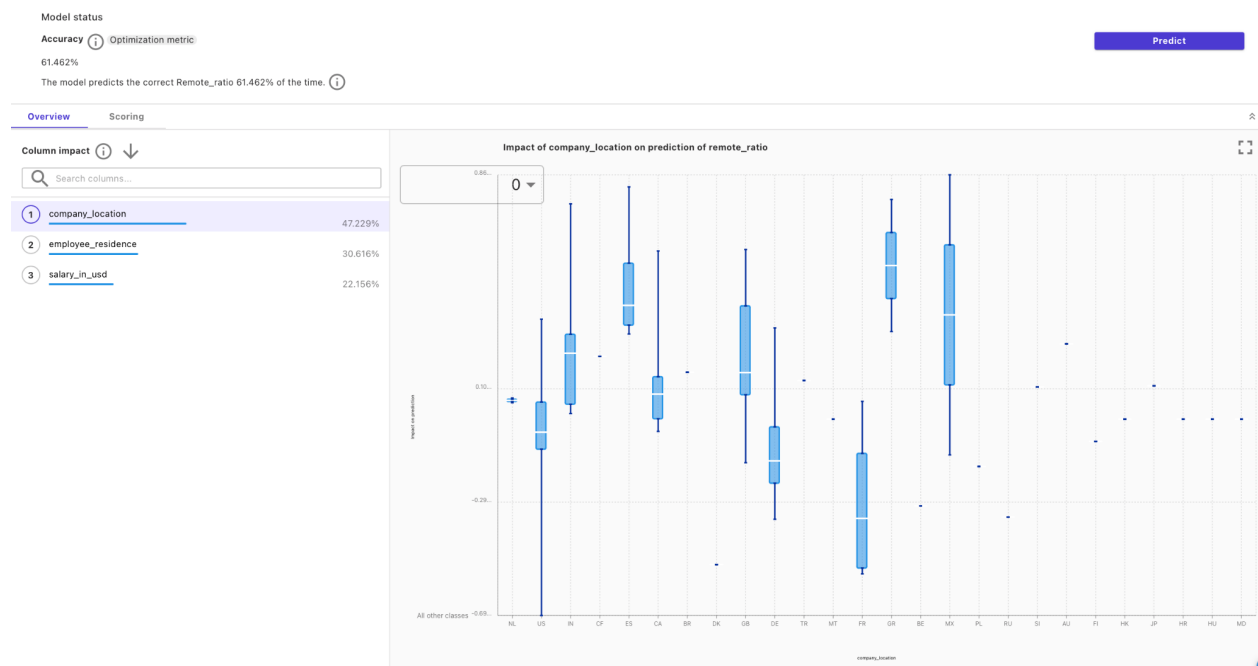


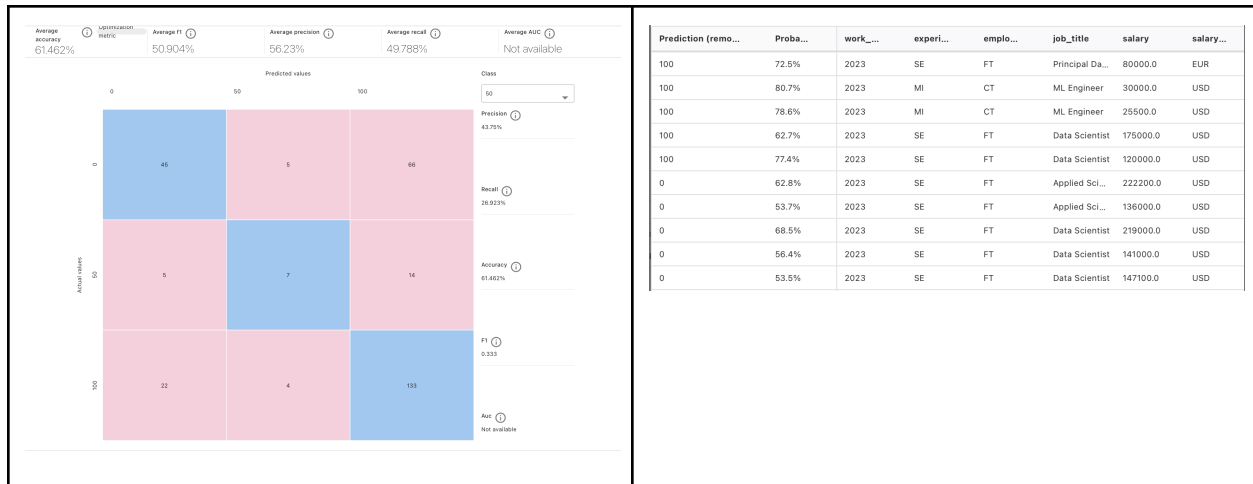
- The 5 factors which would have had correlation to salaries paid are, company location, experience level, job title, remote ratio and company size. Company location as seen in the bar chart above shows salary disparity between different locations. Next is

experience level, this is where depending on how experienced an employee the pay generally goes up, as evidence it can be seen that the 4 different recorded experience levels all have different pays with “EX” getting paid far more. Job title reflects the roles and jobs of the employees and as such different roles have different importance and will be paid accordingly. Another factor is remote ration where individuals who work remotely will have a different pay to those who need to attend a company location. The last factor is company size where as seen in the data exploration section salary vs company size the pay for the different sized companies do in fact pay different with medium sized companies paying the most.

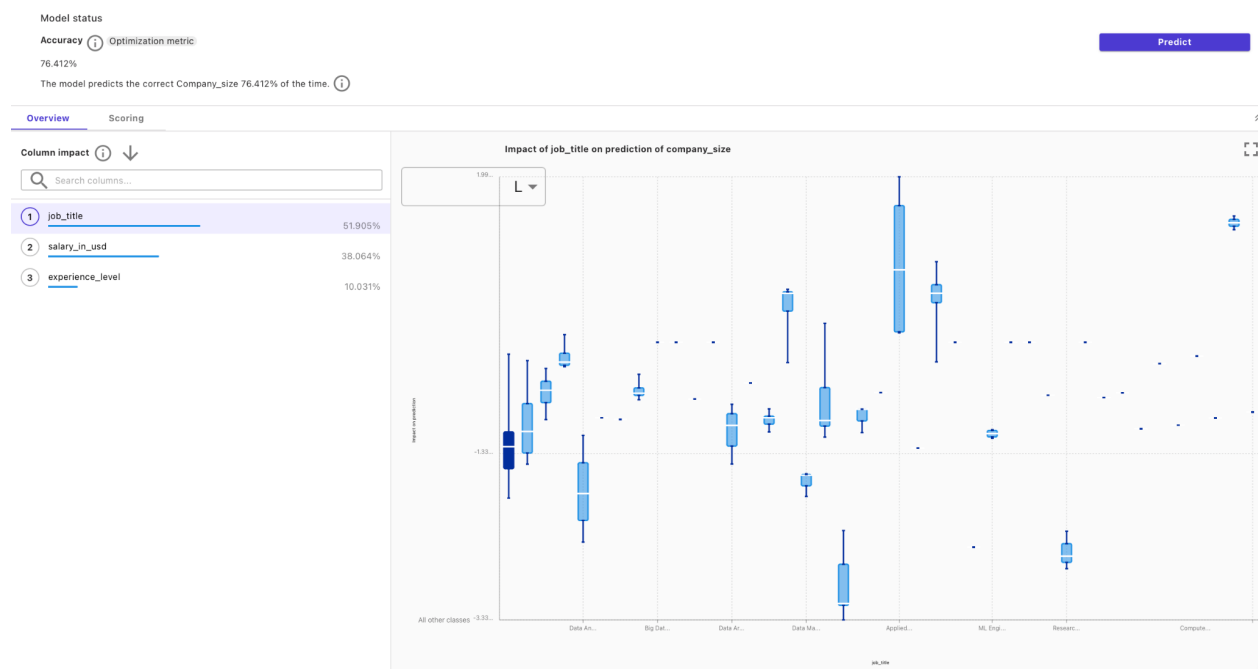
Data modelling

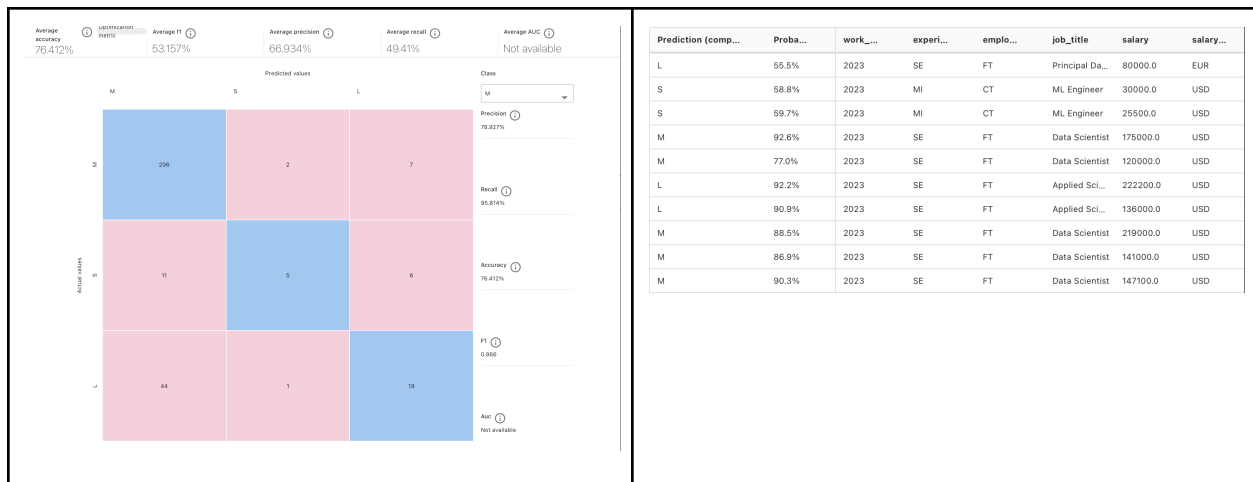
- The model below targets the column Remote_ratio, aiming to predict the ratio of employees who work remotely or at a company location using the classification model. Using the variables company_location, employee_residence and salarin_in_usd we are able to predict to an accuracy of 61.462% of whether an individual given those variables will be working remotely or not. The first image shows the contribution each column has to helping to make the prediction and company_location has the most weight at 47.229%. Below that on the left is an advanced look showing different metrics and measures used to evaluate the performance of models. Looking at which the model used here is not a strong predictive model where there is a high chance of error. To the right we have tested the data using the same “2023-S2-FOAI-assignment.csv” data set, and have displayed the first 10.



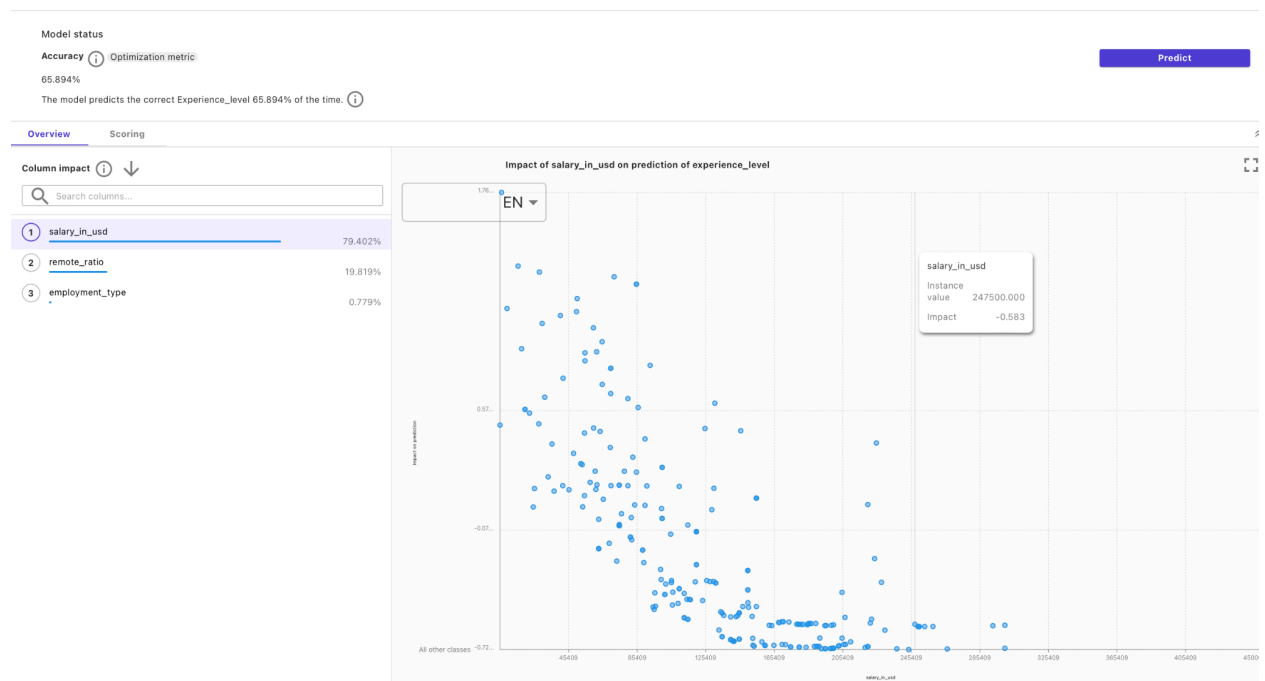


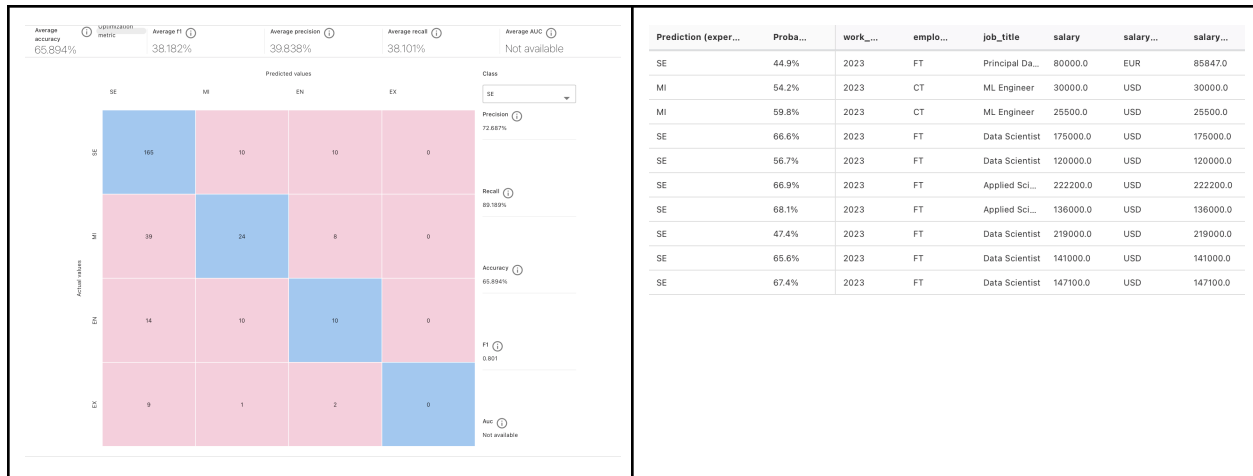
- In this model we looked at company_size based on job_title, salary_in_USD and experince_level. And for this model the accuracy was much higher at 76.412%. Job Title had the most contribution in the prediction, followed by salary_in_usd and then experience_level. The aim of this model was to predict whether the company size would be S,M or L correlating to the size small, medium or large. The data for this model as well as the others are using the cleaned version of “2023-S2-FOAI-assignment.csv” where the missing values and outliers have been removed. This model has mostly met our expectation however it was assumed that salary would be the more contributing factor since it was demonstrated in visualisation that company size and salary were correlated. To improve on the model other variables such as company location could be a better predictive source.





- In the following model the aim was to predict whether an employee was in a junior, mid-level, senior, or executive role. This model can help to assign individuals into their correct and respective experience_levels, this can be useful for career advancement, salary compensation, profiling and much more, this is important to companies. Using a classification model using salary_in_usd, remote_ratio and employment_type we predicted experience level. Salary_in_usd had the greatest impact at 79.402% and employment_type barely contributed at 0.779%. The accuracy of this model is not very high only being at 65.894% this could be due the the fact that the main predictive source was sourced from a single column.





Data Interpretations

In analysing the dataset "2023-S2-FOAI-assignment.csv," we found several key insights, regarding the relationship between various factors in the data science industry, highlighting the relationships between variables that influence salaries, job roles, and work other arrangements within this industry.

- **Differences in salary vs Job Title:** The analysis reveals significant variations in average salaries based on job titles. Roles such as Computer Vision Engineer and analytics experts command higher salaries, while roles like Compliance Data Analyst and ML Engineers tend to have lower compensation.
- **Company Location vs Salary:** The data shows clear salary disparities based on the company's location. Countries like the United Kingdom and the United States tend to offer higher average salaries compared to other locations.
- **Company Size vs Salary:** The relationship between company size and average salary suggests that medium-sized companies provide higher compensation on average. This trend may reflect the balance between resources and financial capabilities.
- **Experience Level vs Salary:** Experience level strongly correlates with salary, with more experienced individuals earning higher pay. This highlights the importance of recognizing and rewarding expertise within the industry.
- **Remote ratio vs salary:** The extent of remote work influences compensation, with employees working 80%+ fully remote potentially earning different salaries compared to those working on-site.

Response to Investigation:

- Through the investigation on the data using modelling and visualisation of the findings achieved our statement and goal of creating a fair and equitable system that accurately looks at rewards for individuals based on their unique circumstances, skills, and roles.
- Our analysis highlights that compensation differences are influenced by a variety of factors where they are intertwined. For example, where someone works (eg, company_location) and what they do (eg, job_titles), have an effect on other aspects which can determine or predict what other circumstances that surround them.

Recommendations:

In conclusion to our data analysis, it is recommended that in order to improve of the fairness and equality of the data science industry we utilise the recognised patterns and predictions in the following means:

- Open Pay Structure: this is where we can establish an open salary framework that factors in location, company size, and experience level in order to determine suitable pay. This in turn should offer employees a clear understanding of how their pay is determined, promoting fairness within the industry and across companies.
- Cross referencing: this is where companies regularly check industry trends and industry salary benchmark and ensure they stay up-to-date. Companies would then be able to ensure their rewards for employees remains competitive and fair within the ever changing job titles and roles.
- Remote_ratio Compensation: This is where we develop systems that determine pay accounting for remote work employees. In short we adjust salary for roles based on the three levels: 0% remote, 50% partial remote, and 80%+ fully remote.
- Skill-Based Rewards: this is where In relation to experience we recognise and compensate employees based on skills and experience level of individuals. We Implement based on factors such as job titles and experience level to fairly determine compensation that separate individuals pay accordingly .

Ultimately, our data analysis serves as a basis for creating a fair rewards system that acknowledges the different contributing factors that influence salary and more in the data science industry in which companies can reference as guidelines for tailored employee work conditions.

References

- Sage maker
- Davenport, T.H. and Patil, D.J. (2017). Data Scientist: The Sexiest Job of the 21st Century. [online] Harvard Business Review. Available at: <https://hbr.org/2012/10/data-scientist-the-sexiest-job-of-the-21st-century>